Self-supervised Learning
System-level SSL

Hermann Blum

UNIVERSITÄT BONN

LAMARR INSTITUTE FOR MACHINE LEARNING AND ARTIFICIAL INTELLIGENCE

Robot Perception and Learning Lab
rpl.uni-bonn.de

Exam Question

Will we be asked to derive loss functions?

- There are no mathematical derivations of losses in the exam.

You should be able to

- explain different loss functions and their parts
- choose the appropriate loss for a given problem
- spot mistakes
- implement losses

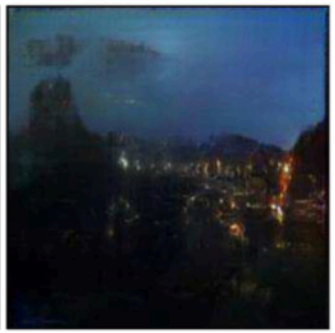
Exam Rooms

The exam on Aug 4 will take place in Hörsaalzentrum in lecture halls 1 and 2.

This is somehow not yet visible in BASIS.

Paired and Unpaired Data

paired data used in conditional generation



strictly, paired data is rather semi-supervised

we might consider them SSL if we can mine pairs without any manual effort, e.g. super-resolution, augmentations

What if data pairs are collected together?

most world-observing devices have ‚rigs‘ of sensors

What if data pairs are collected together?

On a **system level**, self-supervised can be interpreted as learning from all available data of the system, but without any external input or labelling.

System-level Pseudo Labels

cs.RO] 20 Aug 2022

Continual Adaptation of Semantic Segmentation using Complementary 2D-3D Data Representations

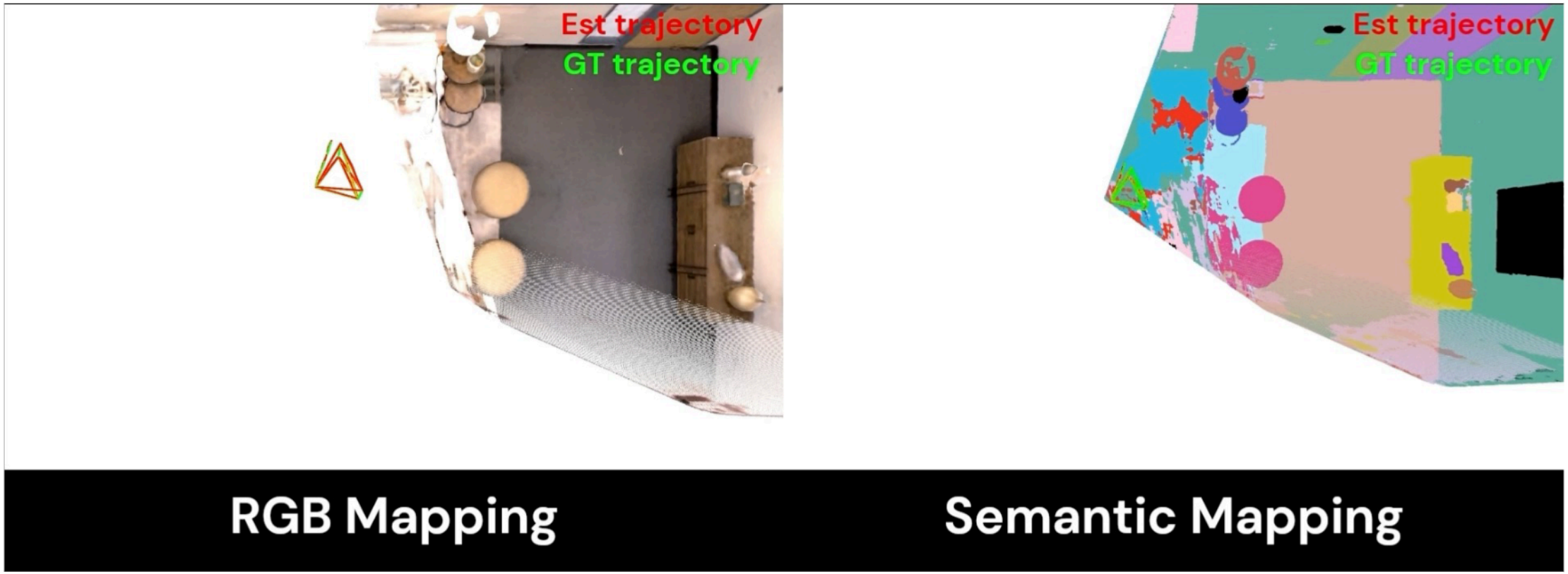
Jonas Frey[1], Hermann Blum[2], Francesco Milano[2], Roland Siegwart[2], Cesar Cadena[2]

***Abstract*— Semantic segmentation networks are usually pre-trained once and not updated during deployment. As a consequence, misclassifications commonly occur if the distribution of the training data deviates from the one encountered during the robot's operation. We propose to mitigate this problem by adapting the neural network to the robot's environment during deployment, without any need for external supervision. Leveraging complementary data representations, we generate a supervision signal, by probabilistically accumulating consecutive 2D semantic predictions in a volumetric 3D map. We then train the network on renderings of the accumulated semantic map, effectively resolving ambiguities and enforcing multi-view consistency through the 3D representation. In contrast to scene adaptation methods, we aim to retain the previously-learned knowledge, and therefore employ a continual learning experience replay strategy to adapt the network. Through extensive experimental evaluation, we show successful adaptation to real-world indoor scenes both on the ScanNet dataset and on in-house data recorded with an RGB-D sensor. Our method increases the segmentation accuracy on average by 9.9% compared to the fixed pre-trained neural network, while**

Fig. 1: Comparison of semantic segmentation performance on a recording from an indoor scene. From top to bottom: Pre-trained network predictions (1-Pred), rendered pseudo-

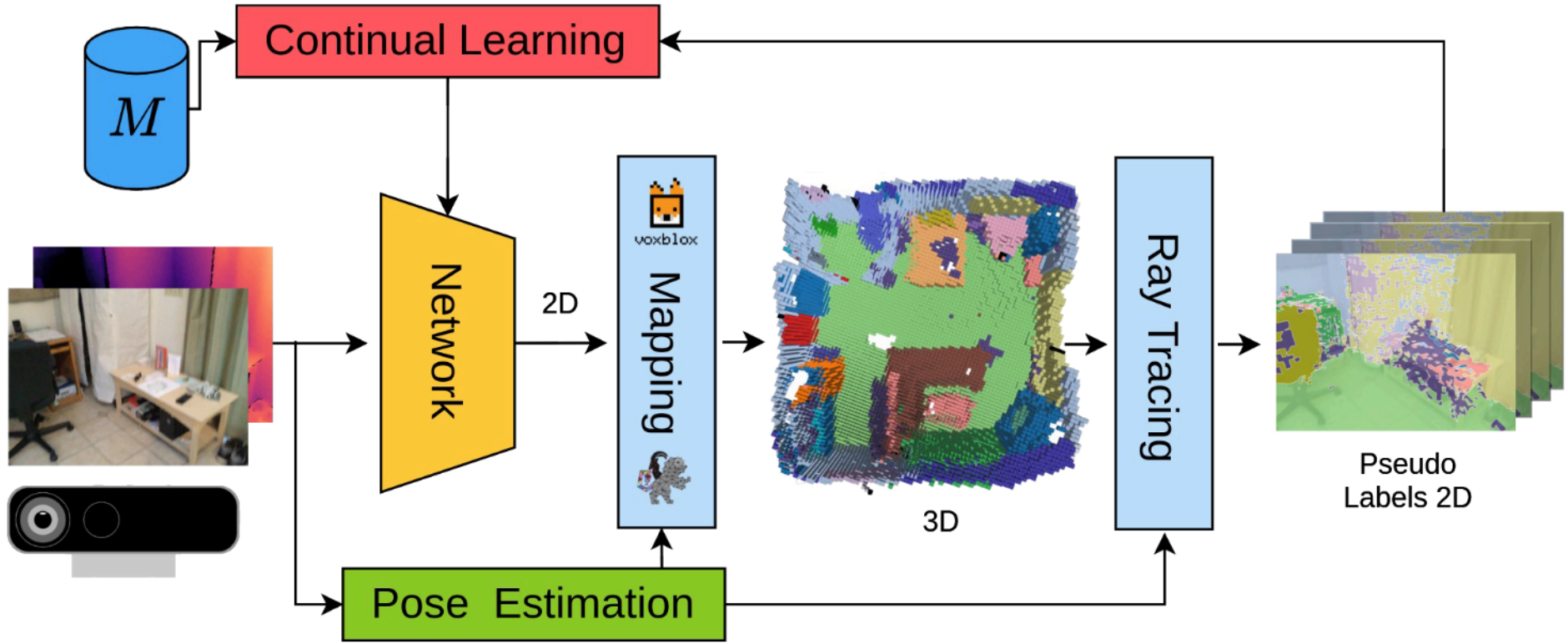
Mapping integrates information from multiple views.

Mapping integrates information from multiple views.



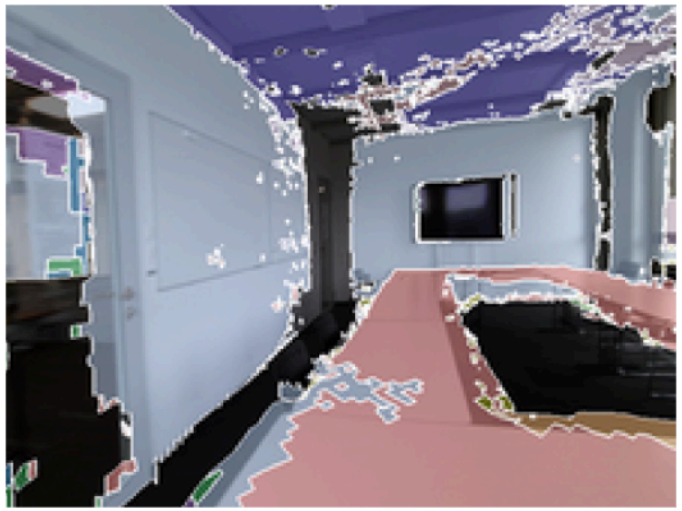
Zhu et al, SNI-SLAM++

Generate pseudo labels by redering from a map.

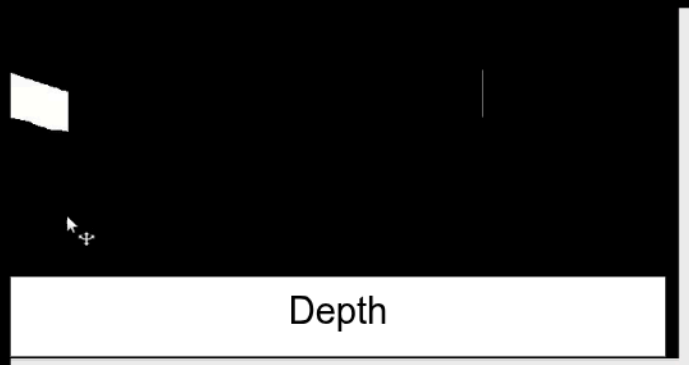


raw teacher prediction

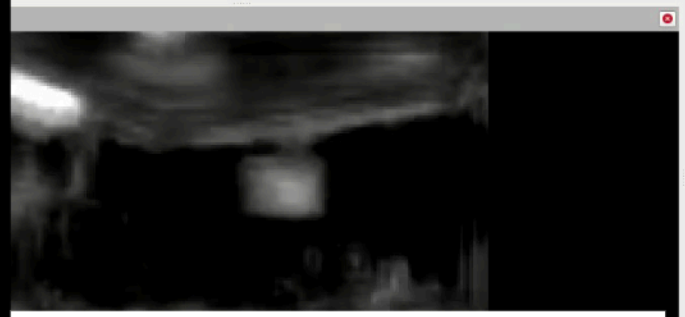
pseudo label = rendered map



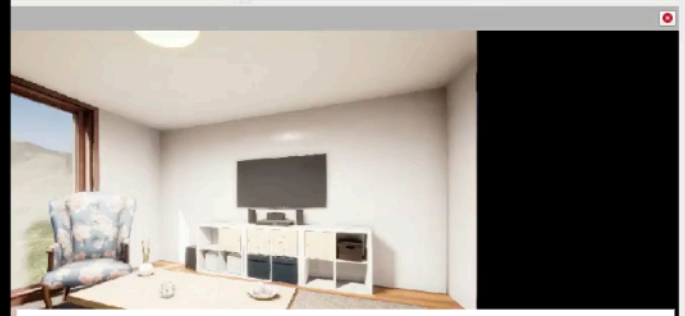
adapted student



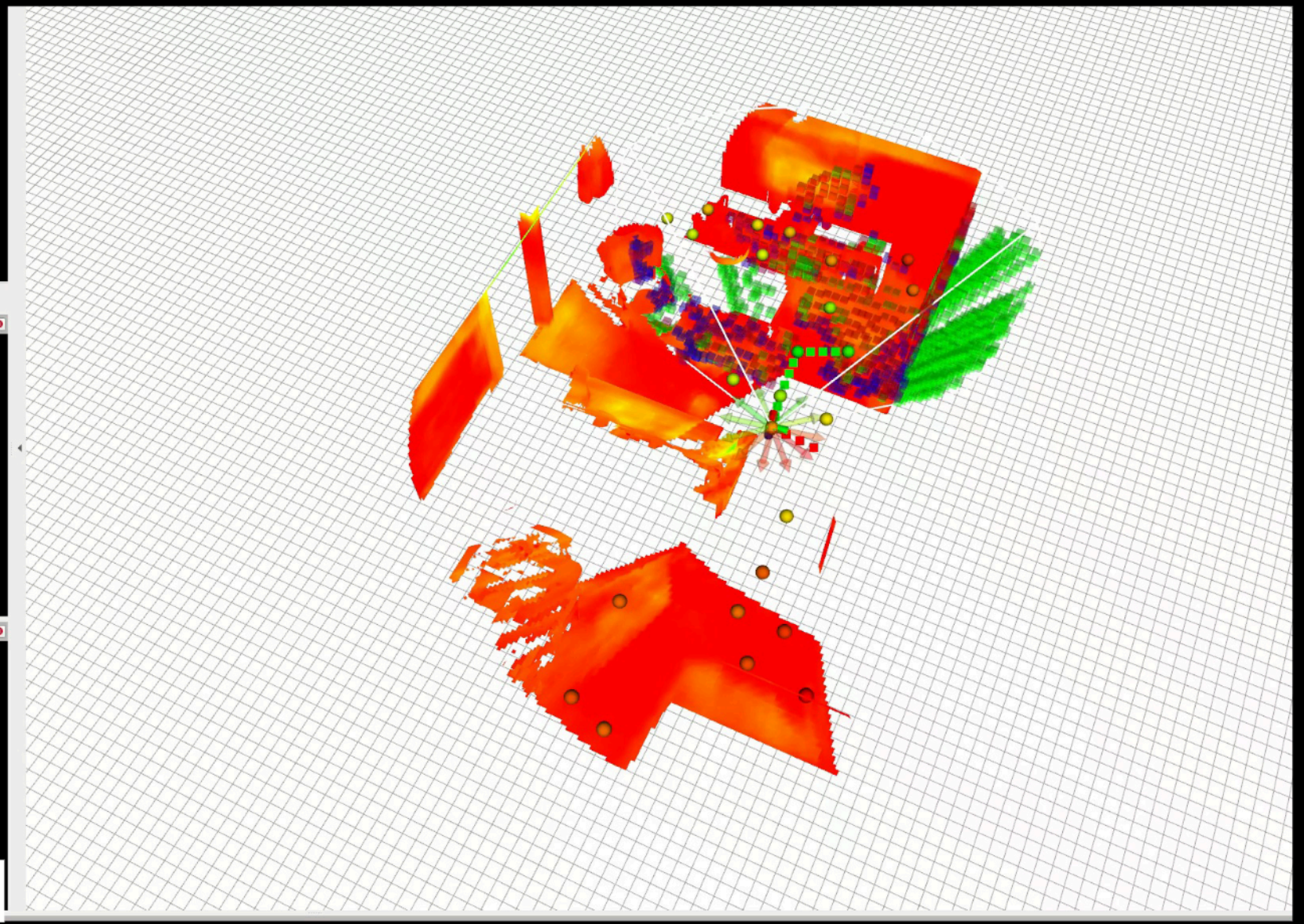
Depth



Uncertainty

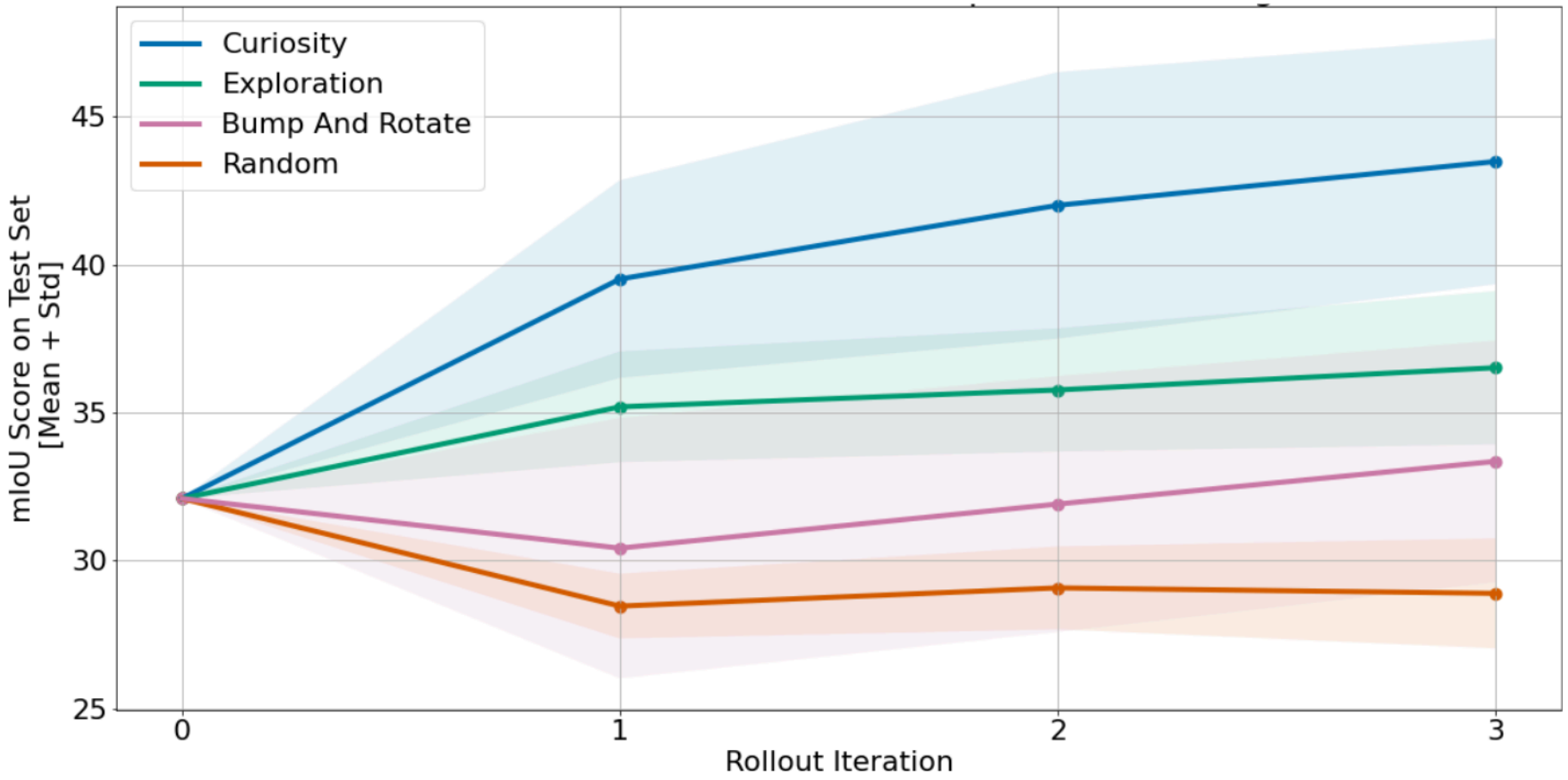


Color



Speed x3

Planning significantly influences adaptation.



Cross-Modal Consistency

This ECCV 2018 paper, provided here by the Computer Vision Foundation, is the author-created version.
The content of this paper is identical to the content of the officially published ECCV 2018
LNCS version of the paper as available on SpringerLink: https://link.springer.com/conference/eccv

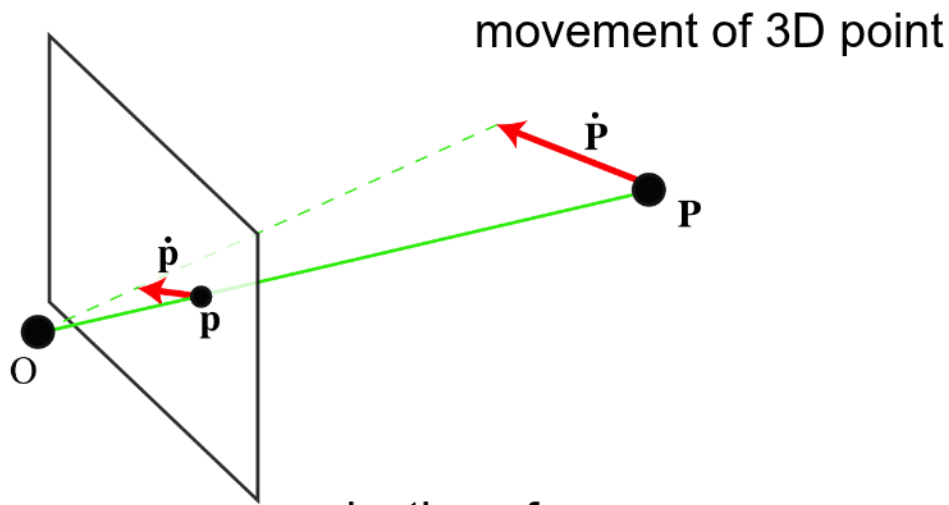
DF-Net: Unsupervised Joint Learning of Depth and Flow using Cross-Task Consistency

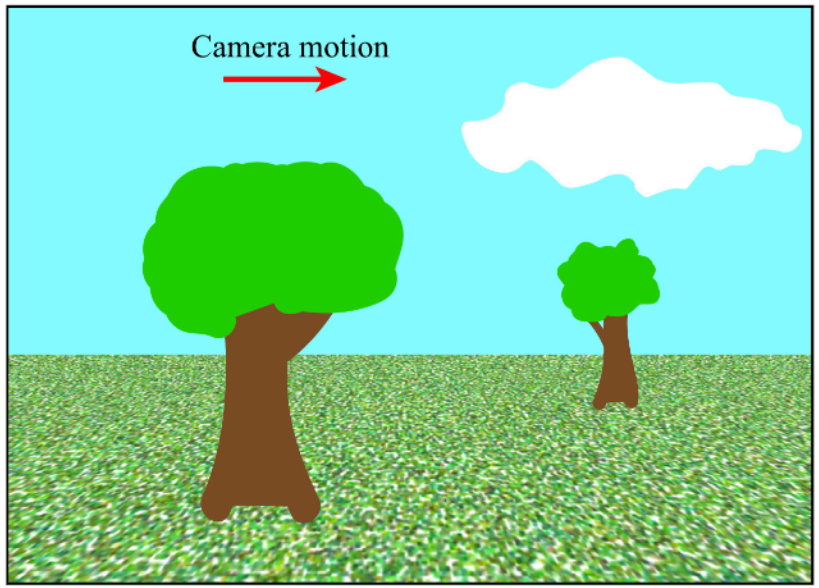
Yuliang Zou[1][0000−0002−8374−6662], Zelun Luo[2][0000−0003−3597−5046], and Jia-Bin Huang[1][0000−0002−0536−3658]

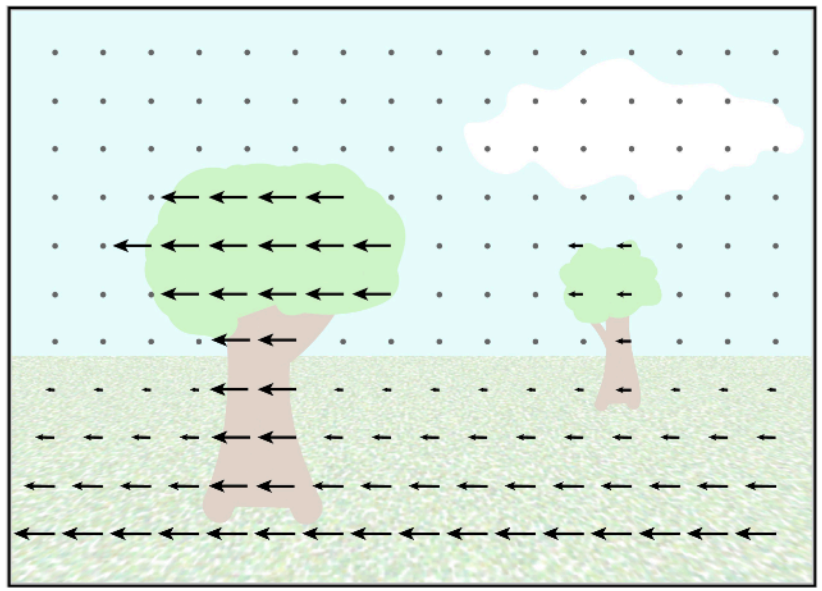
[1]Virginia Tech [2]Stanford University

Fig. 1: **Joint learning v.s. separate learning.** Single-view depth prediction and optical flow estimation are two highly correlated tasks. Existing work,

Optical Flow

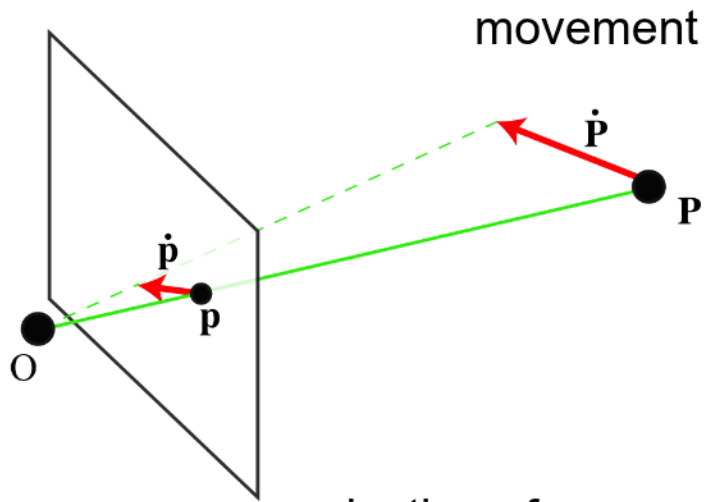


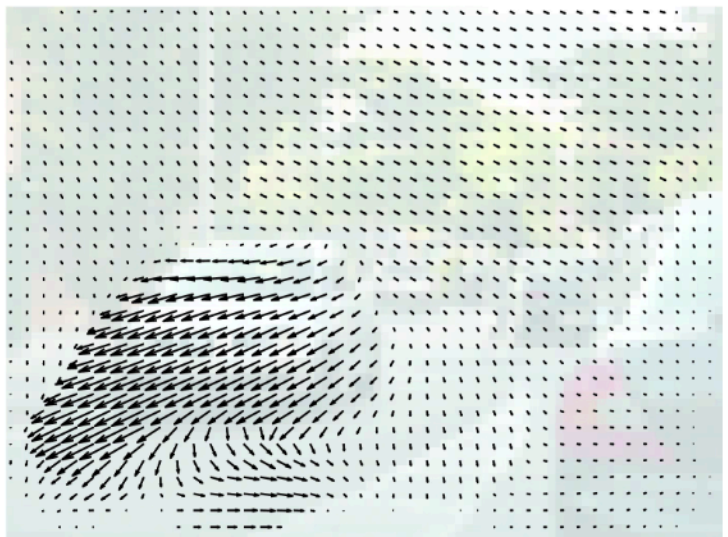




Torralba et al, „Foundations of Computer Vision“

Optical Flow

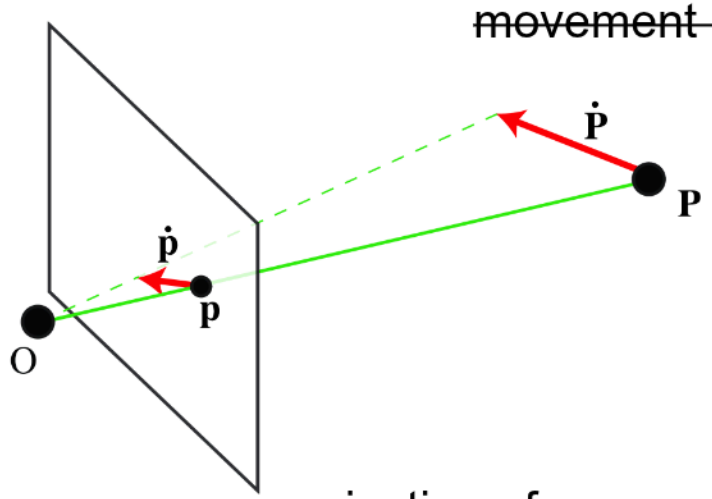




Torralba et al, „Foundations of Computer Vision“

Cross-Modal Consistency

distance to camera of
~~movement of~~ 3D point



movement of
camera origin

projection of
3D point on
image plane

key idea:
learn each by aligning to the 2 others

Flow from Depth and Pose

assuming a static scene, projection yields pixel correspondences

Flow from Depth and Pose

Non-static points can be identified by comparing Frame B with the projection of Frame A and vice-versa

Input frames

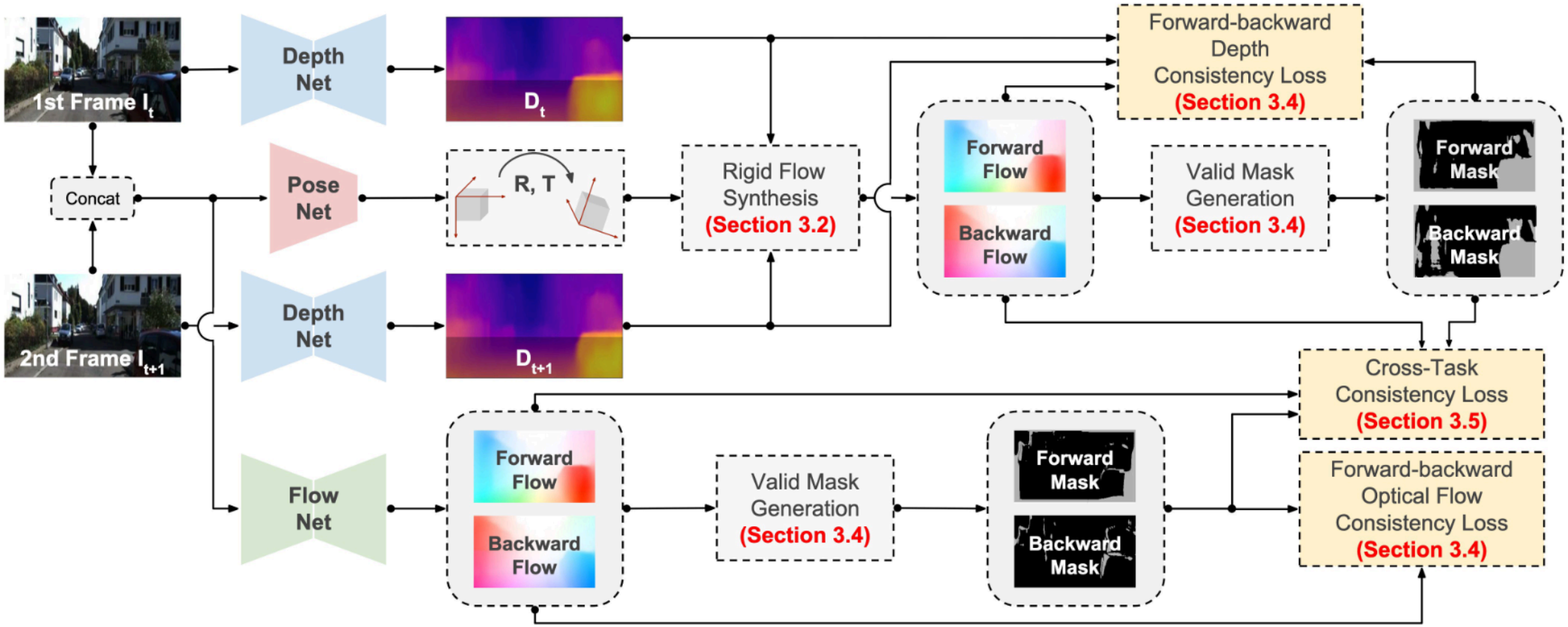
Invalid masks by rigid flow

Forward-Backward-Consistency

depth of frame A = relative pose * depth of frame B

depth of frame B = relative pose * depth of frame A

DF-NET


1st Frame I_t
Depth Net
D_t
Forward-backward Depth Consistency Loss (Section 3.4)
Concat
Pose Net
R, T
Rigid Flow Synthesis (Section 3.2)
Forward Flow
Backward Flow
Valid Mask Generation (Section 3.4)
Forward Mask
Backward Mask
2nd Frame I_{t+1}
Depth Net
D_{t+1}
Cross-Task Consistency Loss (Section 3.5)
Flow Net
Forward Flow
Backward Flow
Valid Mask Generation (Section 3.4)
Forward Mask
Backward Mask
Forward-backward Optical Flow Consistency Loss (Section 3.4)

Full Loss

rendering one frame from the other

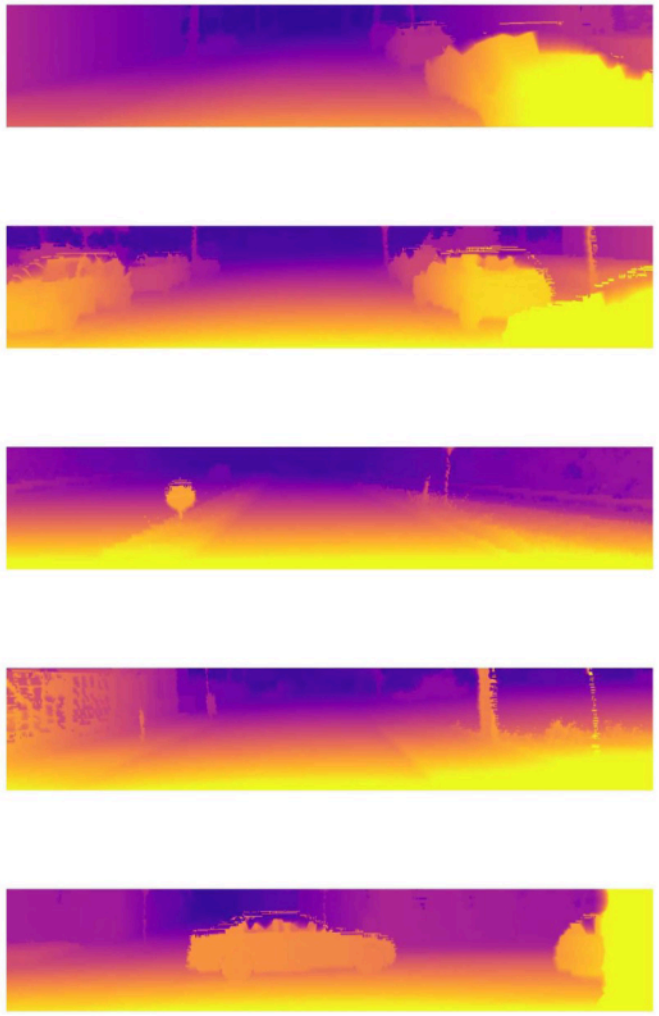
A -> B consistent with B -> A

$$L = L_{\text{photometric}} + \lambda_s L_{\text{smooth}} + \lambda_f L_{\text{forward-backward}} + \lambda_c L_{\text{cross}}.$$

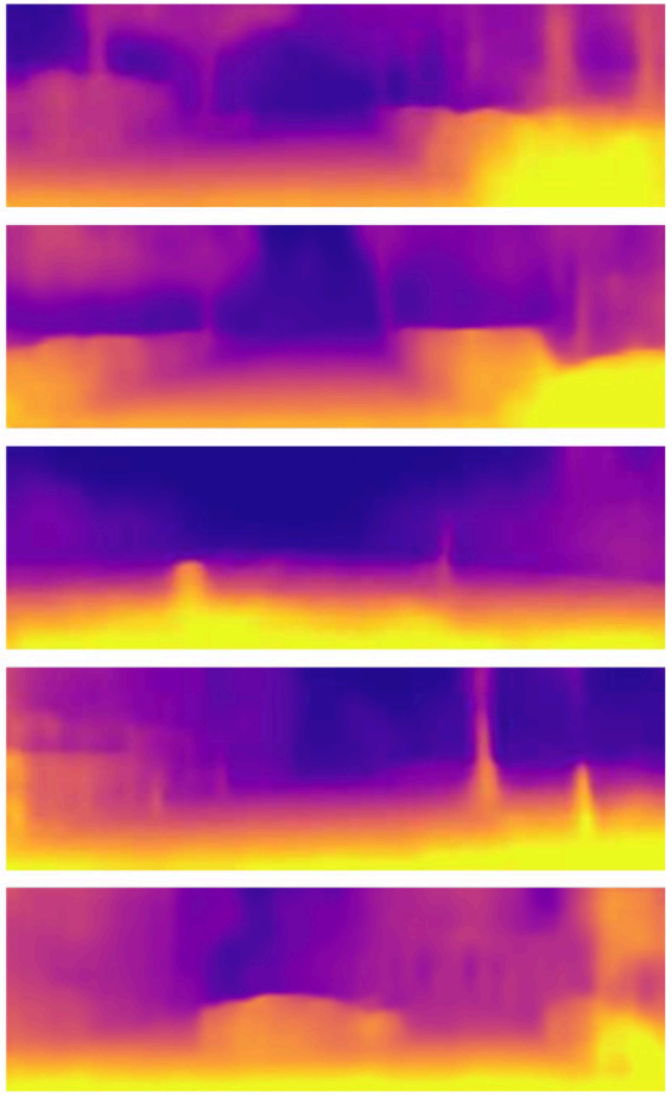
smoothness constraint

flow consistent with depth

Input



Ground truth



Ours

Cross-Modal Estimation

510v2 [cs.RO] 16 May 2023

Fast Traversability Estimation for Wild Visual Navigation

Jonas Frey*,[1,3] Matias Mattamala*,[2] Nived Chebrolu[2] Cesar Cadena[1] Maurice Fallon[2] Marco Hutter[1]
[1] ETH Zurich [2] University of Oxford [3] Max Planck Institute for Intelligent Systems
* Equal contribution. jonfrey@ethz.ch, matias@robots.ox.ac.uk

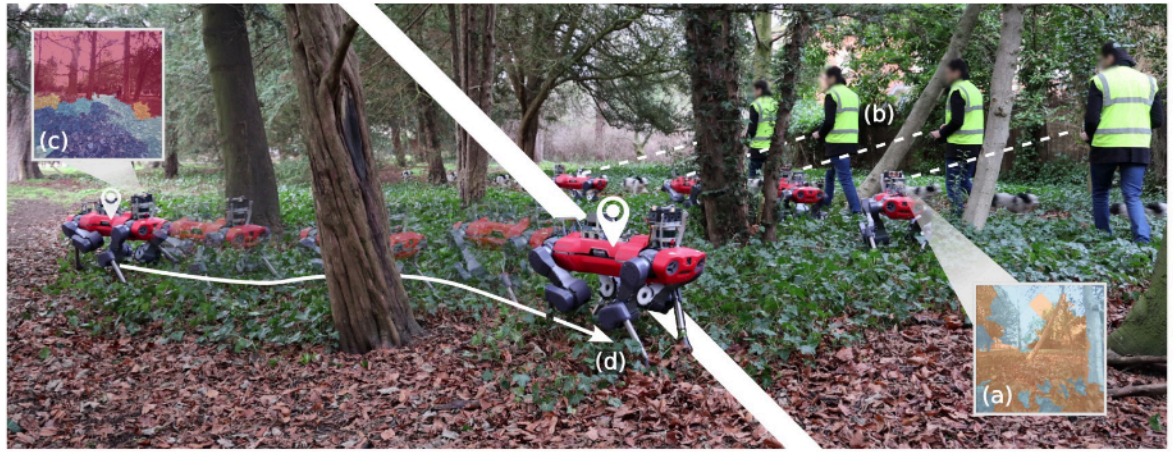
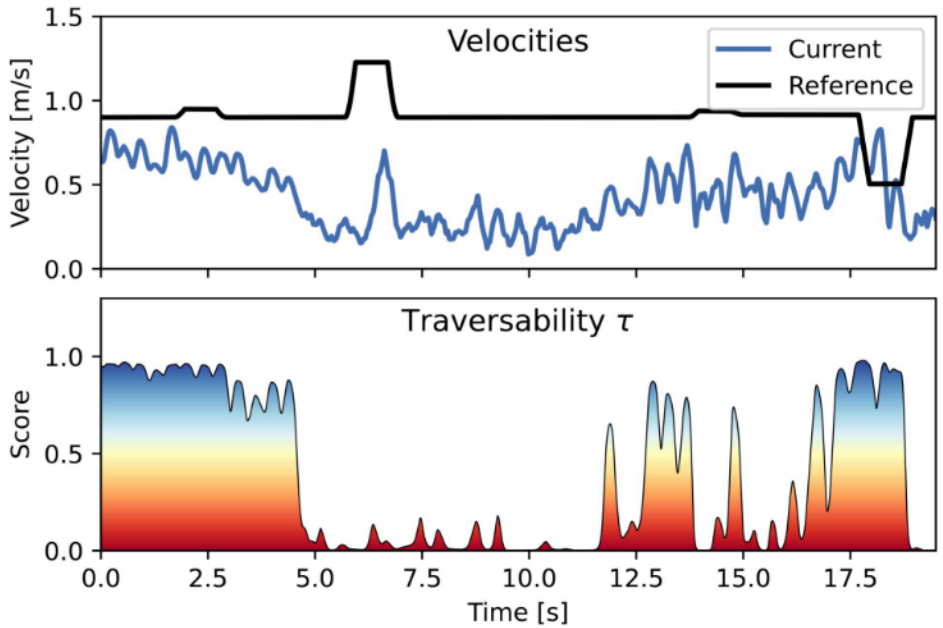


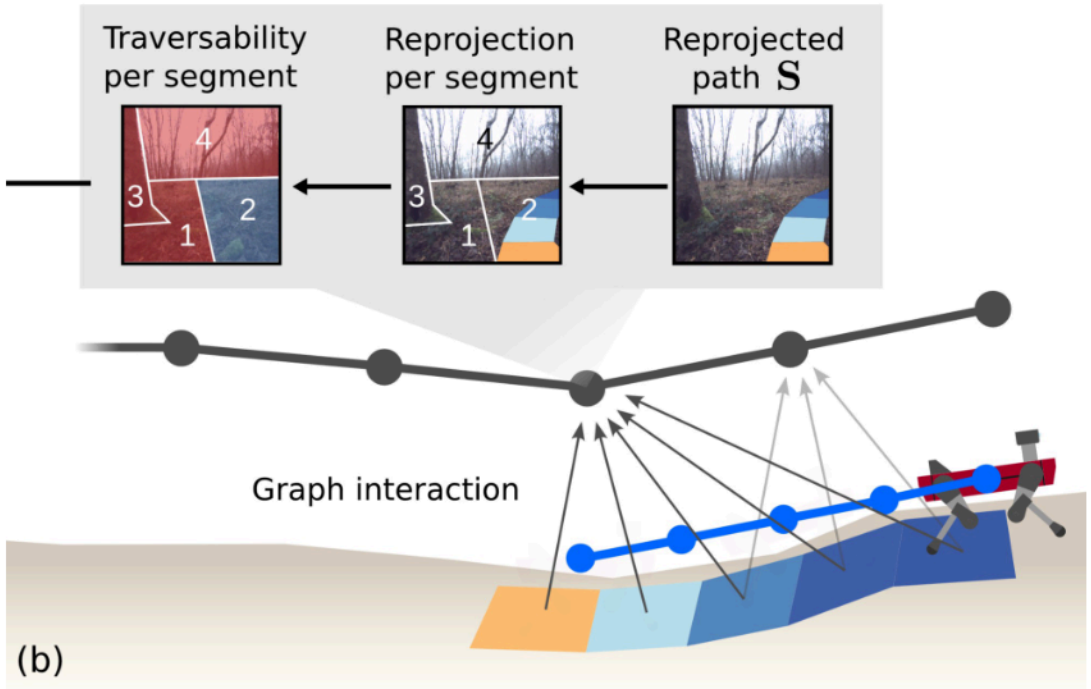
Fig. 1: Wild Visual Navigation (WVN) learns to predict traversability from images via online self-supervised learning. Starting from a randomly initialized traversability estimation network without prior assumptions about the environment (a), a human operator drives the robot around areas that are traversable for the given platform (b). After a few minutes of operation, WVN learns to distinguish between traversable and untraversable areas (c), enabling the robot to navigate autonomously and safely within the environment (d).

generate pseudo-labels from measured experiences

measure traversability as match between target and actual velocity



annotate collected images by reprojection of traversed ground segments



How do you learn where you did not go?

traversability scores

we know nothing

How do you learn where you did not go?

we know nothing

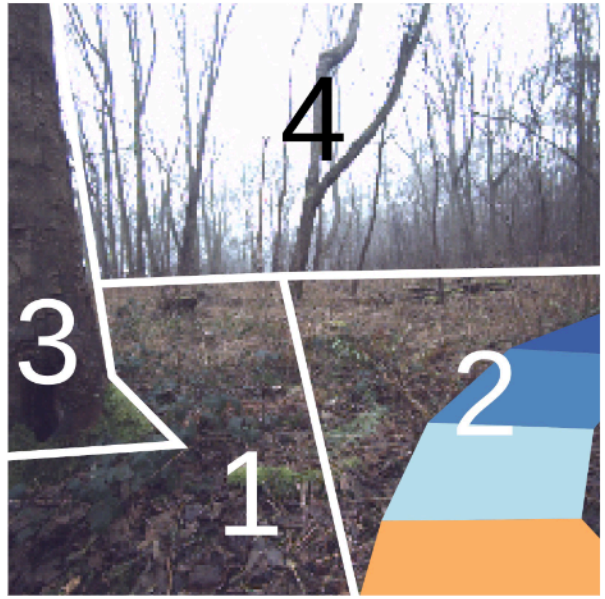
traversability scores

1. extract DINO feature per segment
2. learn auto-encoder on annotated segments
3. identify unknown terrain by reconstruction error

$$\mathcal{L}_{\text{reco}}(\mathbf{f}_n) = \delta_{\tau_n \neq 0} \frac{1}{E} \sum_e \| f_{\text{reco}}^{\theta_r}(\mathbf{f}_{n,e}) - \mathbf{f}_{n,e} \|^2$$

only where we have data

avoid confusing missing annotation for non-traversible terrain



assume 0 traversibility on strange looking terrain

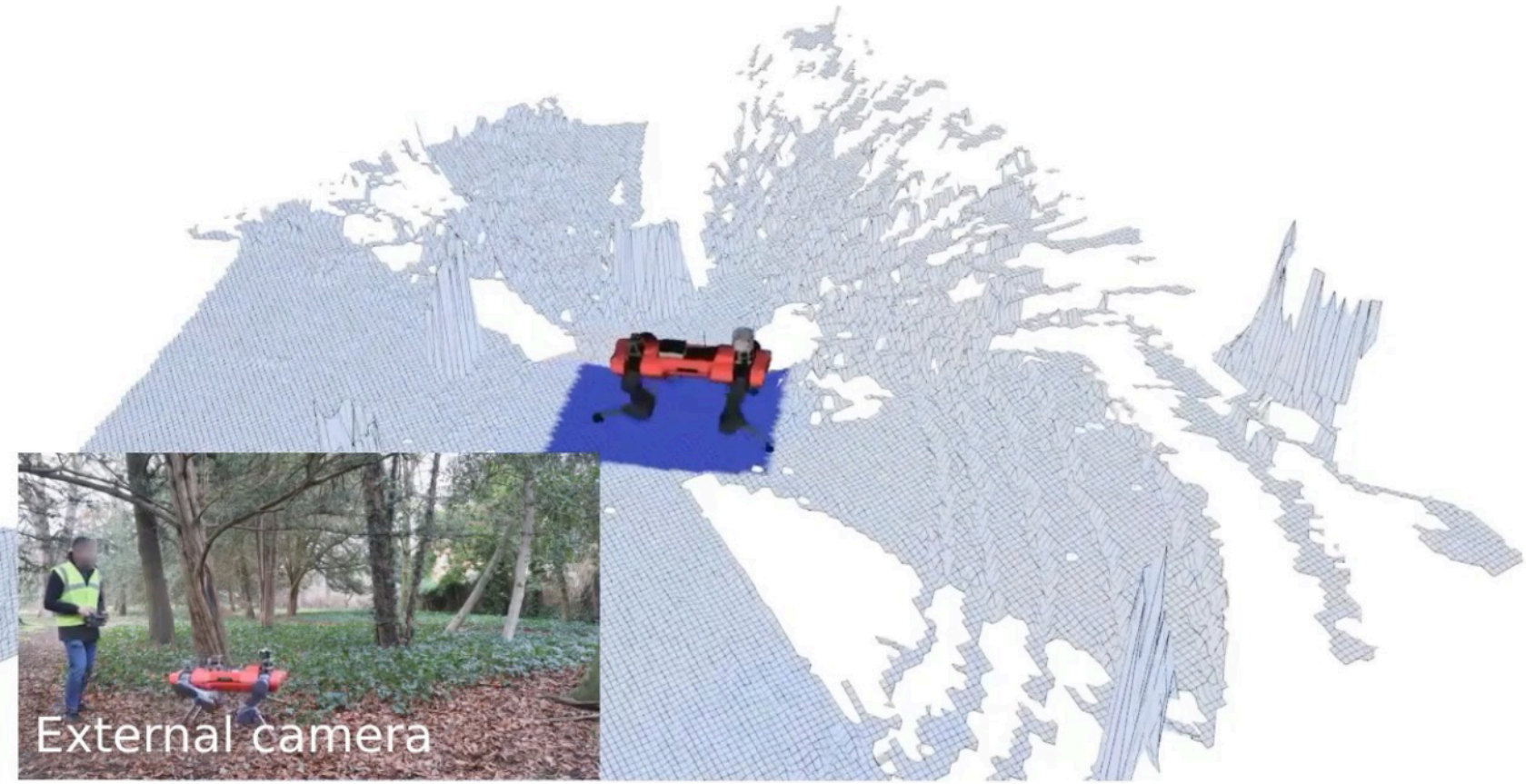
0 for high confidence, i.e. similar features as annotated somewhere else

$$\mathcal{L}_{\text{trav}}(\mathbf{f}) = \delta_{\tau_n=0} \sum_n \overbrace{(1 - c(\mathbf{f}_n))} \| f_{\text{trav}}^{\theta_t}(\mathbf{f}_n) - 0 \|^2 +$$

$$\delta_{\tau_n \neq 0} \sum_n \| f_{\text{trav}}^{\theta_t}(\mathbf{f}_n) - \tau_n \|^2 .$$

for annotated segments, predict the pseudo label

Wild Visual Navigation (WVN) learns traversability online, in a self-supervised way, from images and demonstrations in real environments



Cross-Model Distillation

[cs.CV] 1 Mar 2021

There is More than Meets the Eye: Self-Supervised Multi-Object Detection and Tracking with Sound by Distilling Multimodal Knowledge

Francisco Rivera Valverde* Juana Valeria Hurtado* Abhinav Valada
University of Freiburg
{riverav, hurtadoj, valada}@cs.uni-freiburg.de

Abstract

Attributes of sound inherent to objects can provide valuable cues to learn rich representations for object detection and tracking. Furthermore, the co-occurrence of audiovisual events in videos can be exploited to localize objects over the image field by solely monitoring the sound in the environment. Thus far, this has only been feasible in scenarios where the camera is static and for single object detection. Moreover, the robustness of these methods has been limited as they primarily rely on RGB images which are highly susceptible to illumination and weather changes. In this work, we present

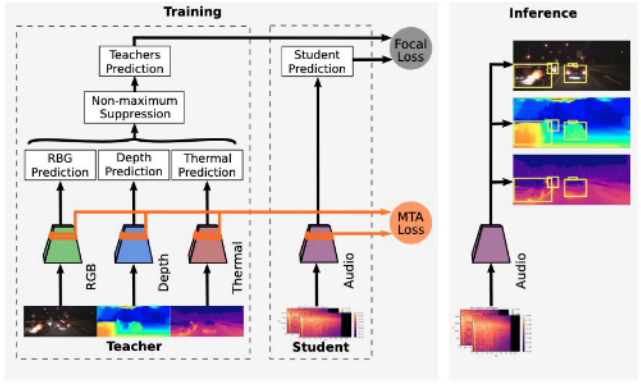
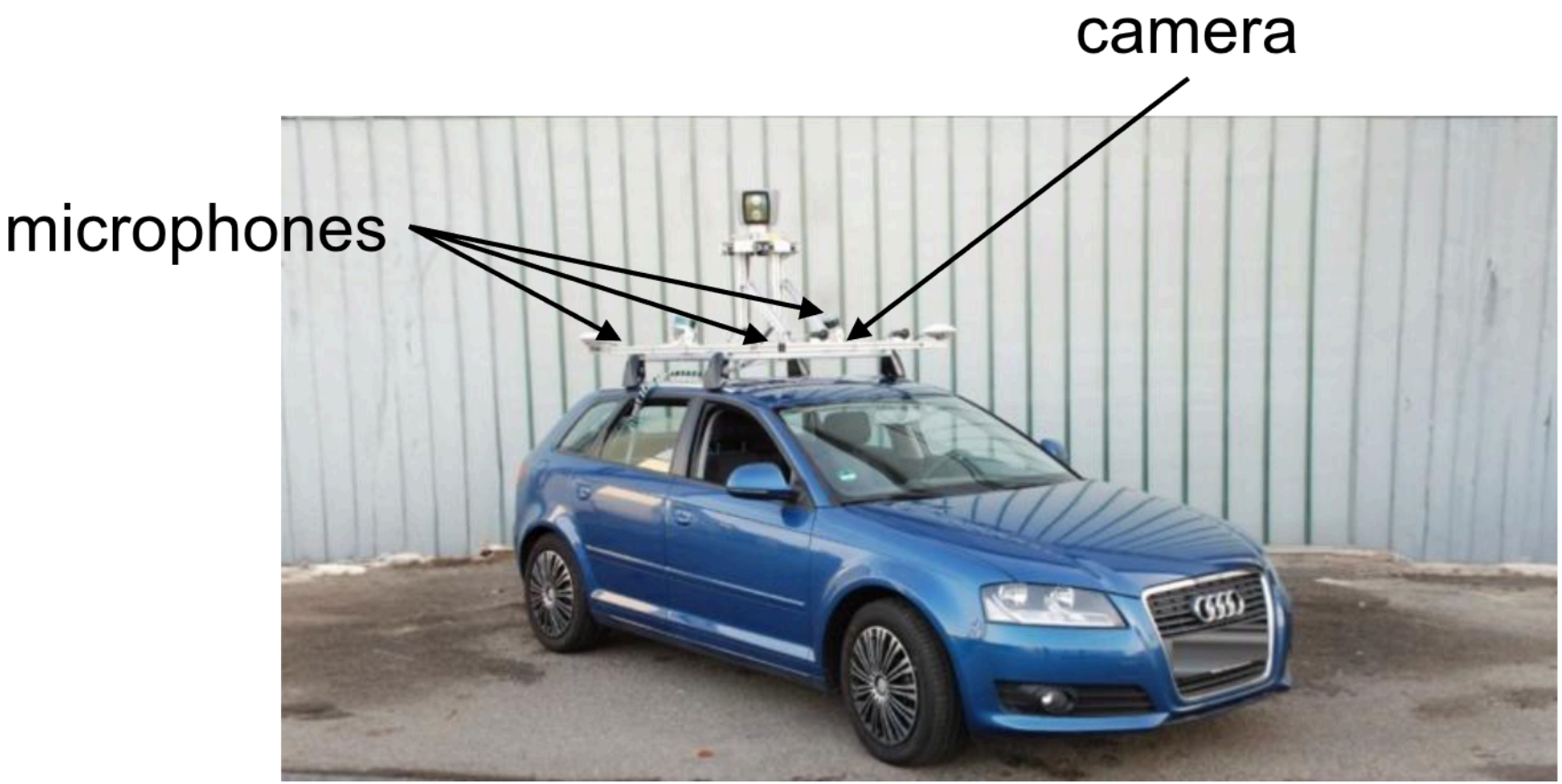
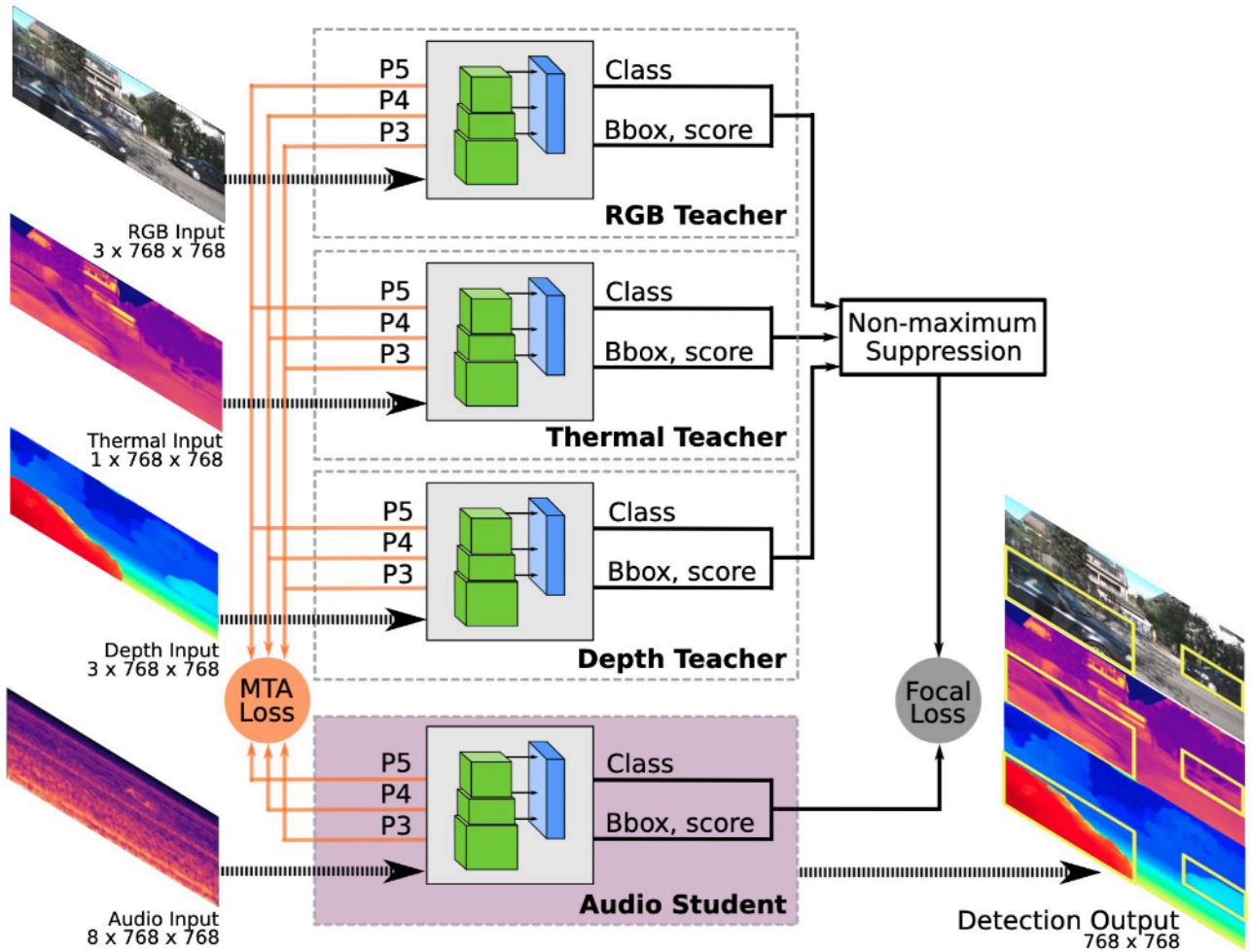


Figure 1. Our proposed cross-modal MM-DistillNet distills knowl-

Learning to process audio from video.



Distillation from multiple teachers



Losses

Focal loss is a variant of cross-entropy

weighting between hard and easy samples

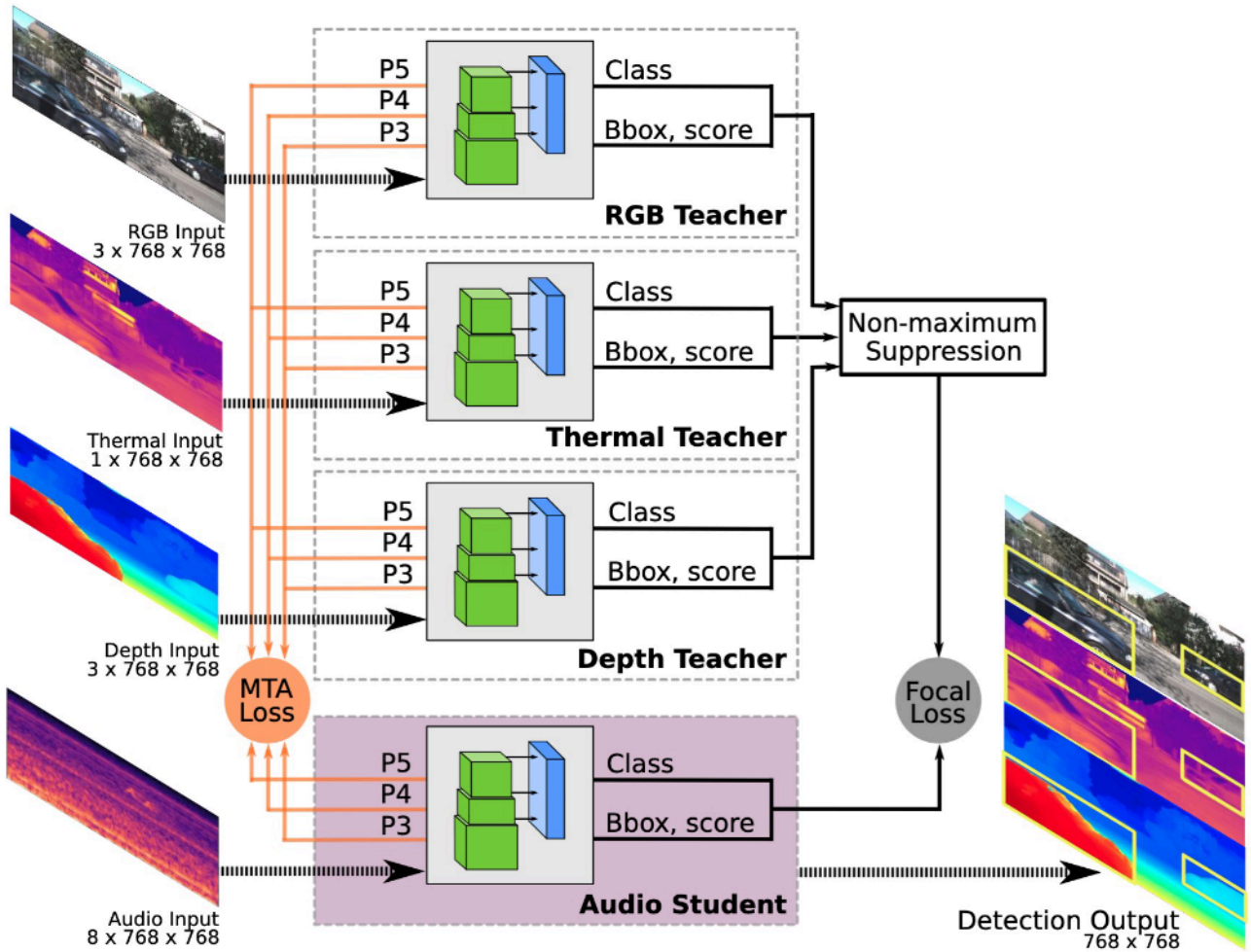
$$L_{focal} = -\alpha\underbrace{(1-pt)^{\gamma}} * log(pt)$$

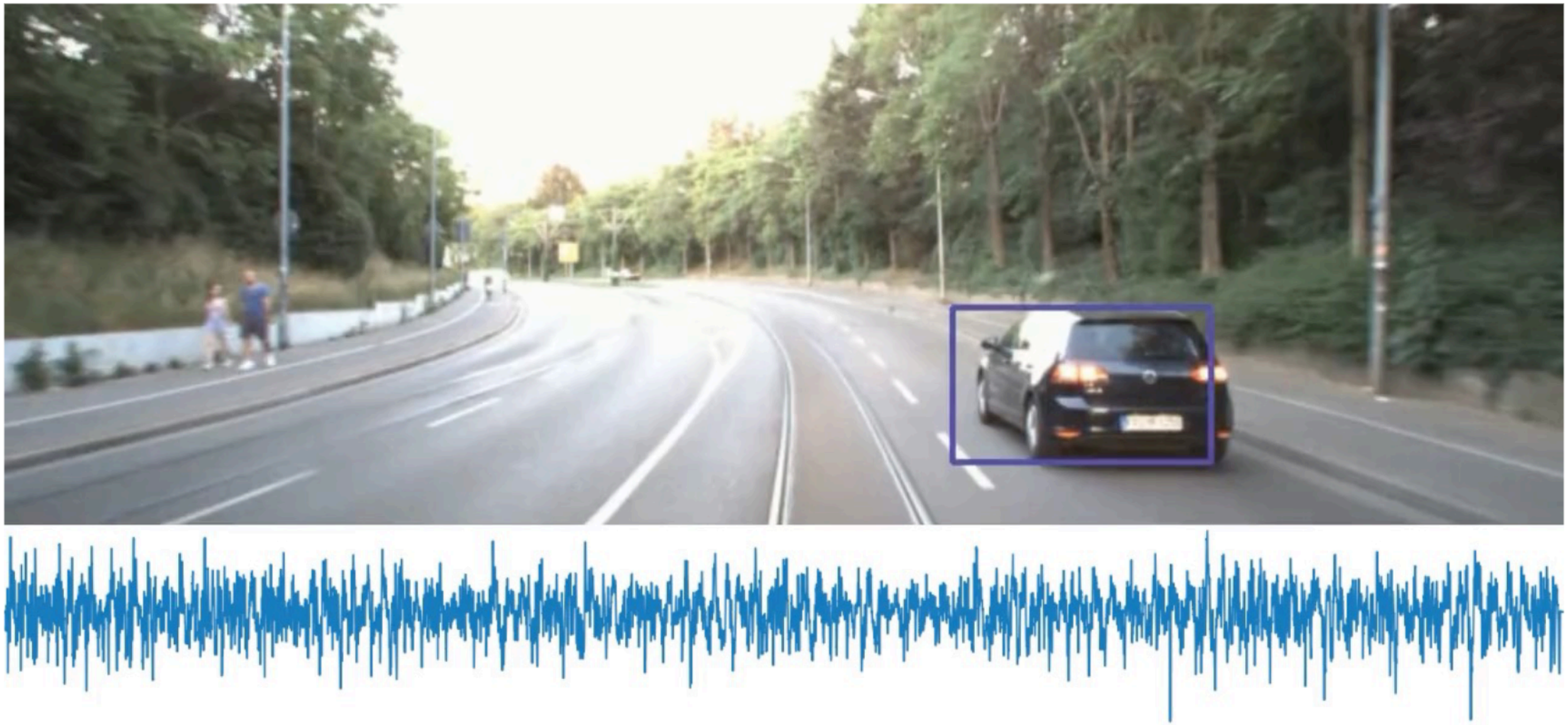
weighting by softmax confidence

divergence between student and teacher attention maps

$$L_{MTA} = \beta * \sum_{j} KL_{div}\left(\frac{Q_s^j}{\left\|Q_s^j\right\|_2}, \frac{Q_t^j}{\left\|Q_t^j\right\|_2}\right),$$

Distillation from multiple teachers





Reinforcement-Learning: Self-supervision by simple reset

Fully Autonomous Real-World Reinforcement Learning with Applications to Mobile Manipulation

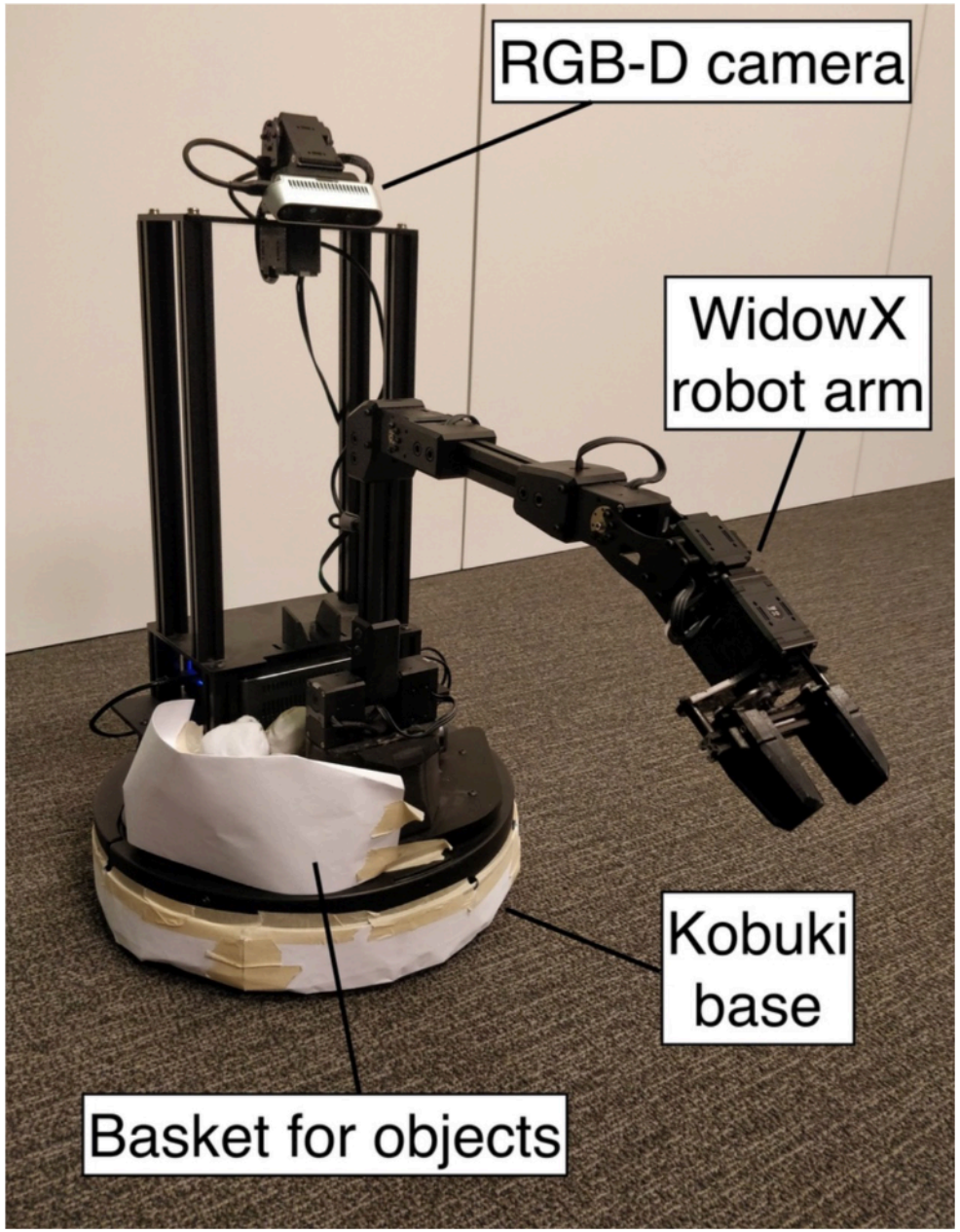
Charles Sun*, **Jędrzej Orbik***, **Coline Devin, Brian Yang,**
Abhishek Gupta, Glen Berseth, Sergey Levine
{charlesjsun,jedrzej.orbik,coline,brianhyang,abhigupta,gberseth,svlevine}@berkeley.edu
Berkeley AI Research

Abstract: We study how robots can autonomously learn skills that require a combination of navigation and grasping. While reinforcement learning in principle provides for automated robotic skill learning, in practice reinforcement learning in the real world is challenging and often requires extensive instrumentation and supervision. Our aim is to devise a robotic reinforcement learning system for learning navigation and manipulation together, in an *autonomous* way without human intervention, enabling continual learning under realistic assumptions. Our proposed system, ReLMM, can learn continuously on a real-world platform without any environment instrumentation, without human intervention, and without

Grasp Learning

Algorithm 1 TrainGrasp($G^1, ..., G^M, \mathcal{D}_g, N, st$)

for t = 0, . . . , N steps **do**
  Get grasp observation $\tilde{o}$.
  Sample $a_g \sim \pi_g(\cdot|\tilde{o})$. // see Equation 2
  Perform grasp a_g, receiving $r_g = 0$ or 1.
  Store $(\tilde{o}, a_g, r_g)$ in $\mathcal{D}_g$.
  Update $G^1, ..., G^M$ on $\mathcal{D}_g$
  if st =True, Drop object if held.
  elseif $r_g = 1$ and st =False, **return** r_g
return 0



Grasp Learning

Algorithm 1 TrainGrasp($G^1, ..., G^M, \mathcal{D}_g, N, st$)

1: **for** t = 0, . . . , N steps **do**
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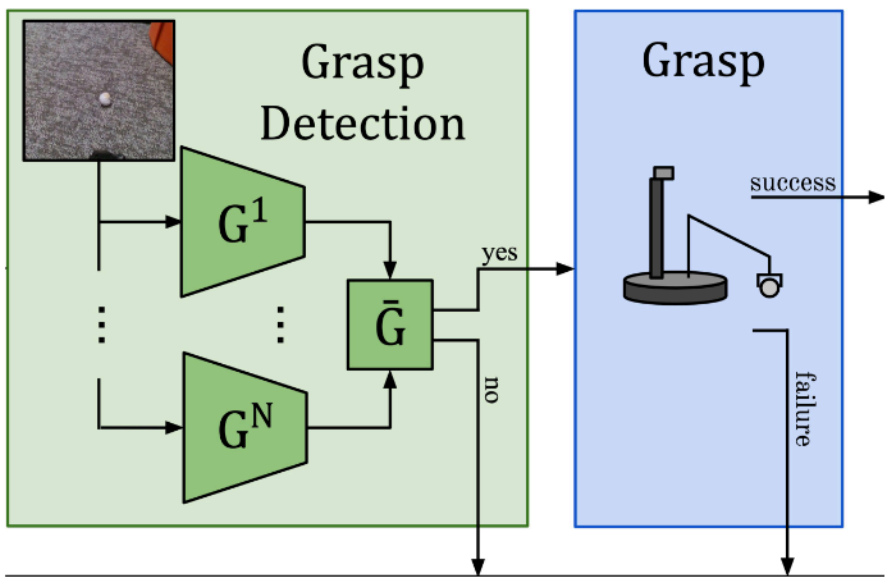
4: Perform grasp a_g, receiving $r_g = 0$ or 1.
5: Store $(\tilde{o}, a_g, r_g)$ in $\mathcal{D}_g$.
6: Update $G^1, ..., G^M$ on $\mathcal{D}_g$
7: **if** st =True, Drop object if held.
8: **elseif** $r_g = 1$ and st =False, **return** r_g
9: **return** 0

sampling trades off high predicted success and high uncertainty

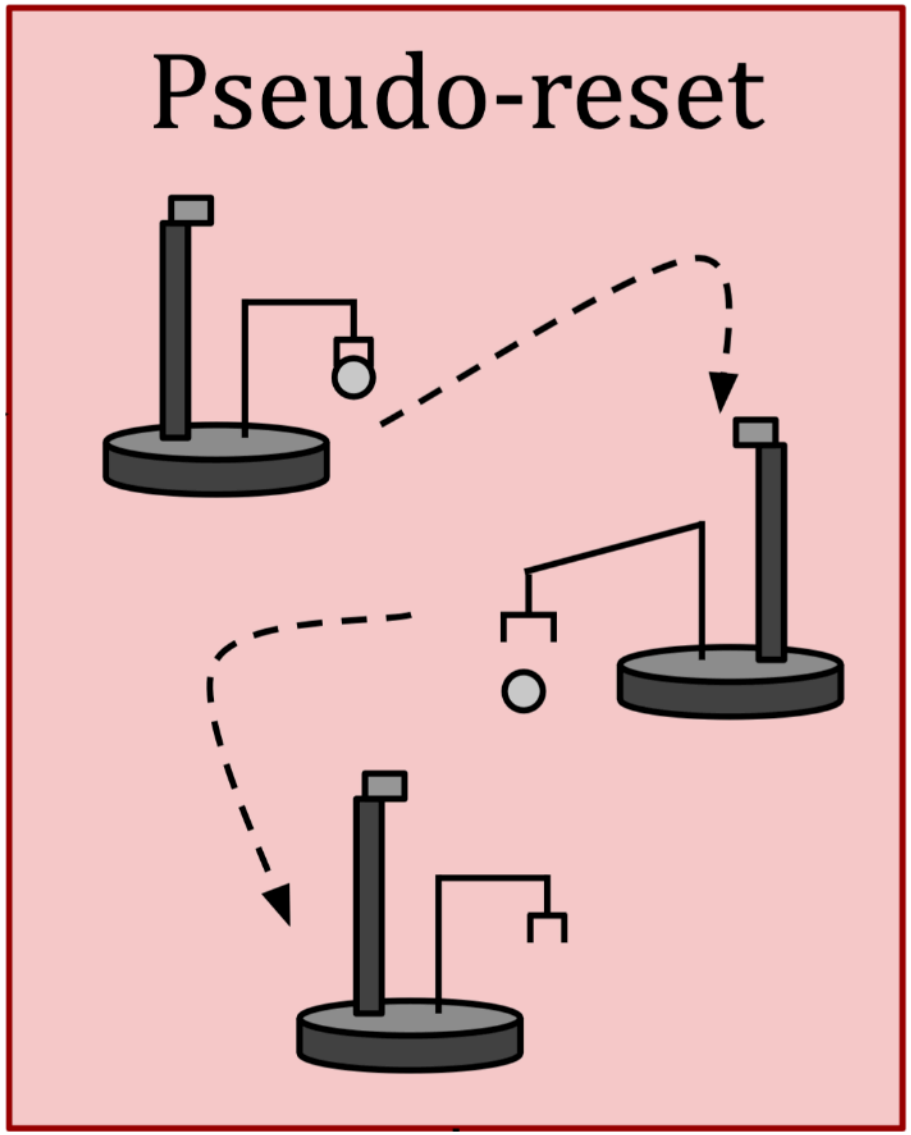
$$\tilde{G}(o_g, a_g) = \alpha \mathbb{E}[G^i(o_g, a_g)] + \beta \sigma(G^i(o_g, a_g)),$$



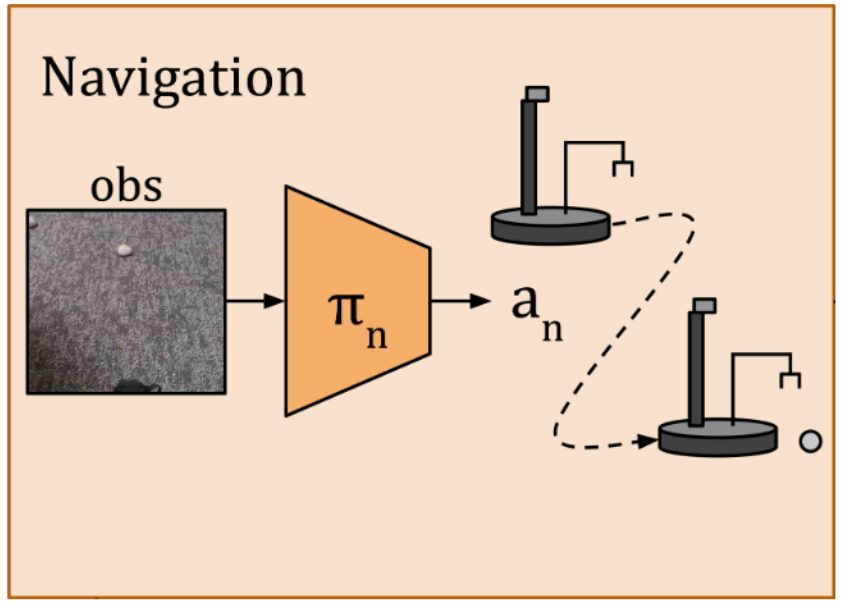
The loss here is simply cross-entropy between predicted grasp and outcome

autonomous reset

1. verify if object in hand
2. drive random path
3. drop object
4. drive random path



Navigation Learning

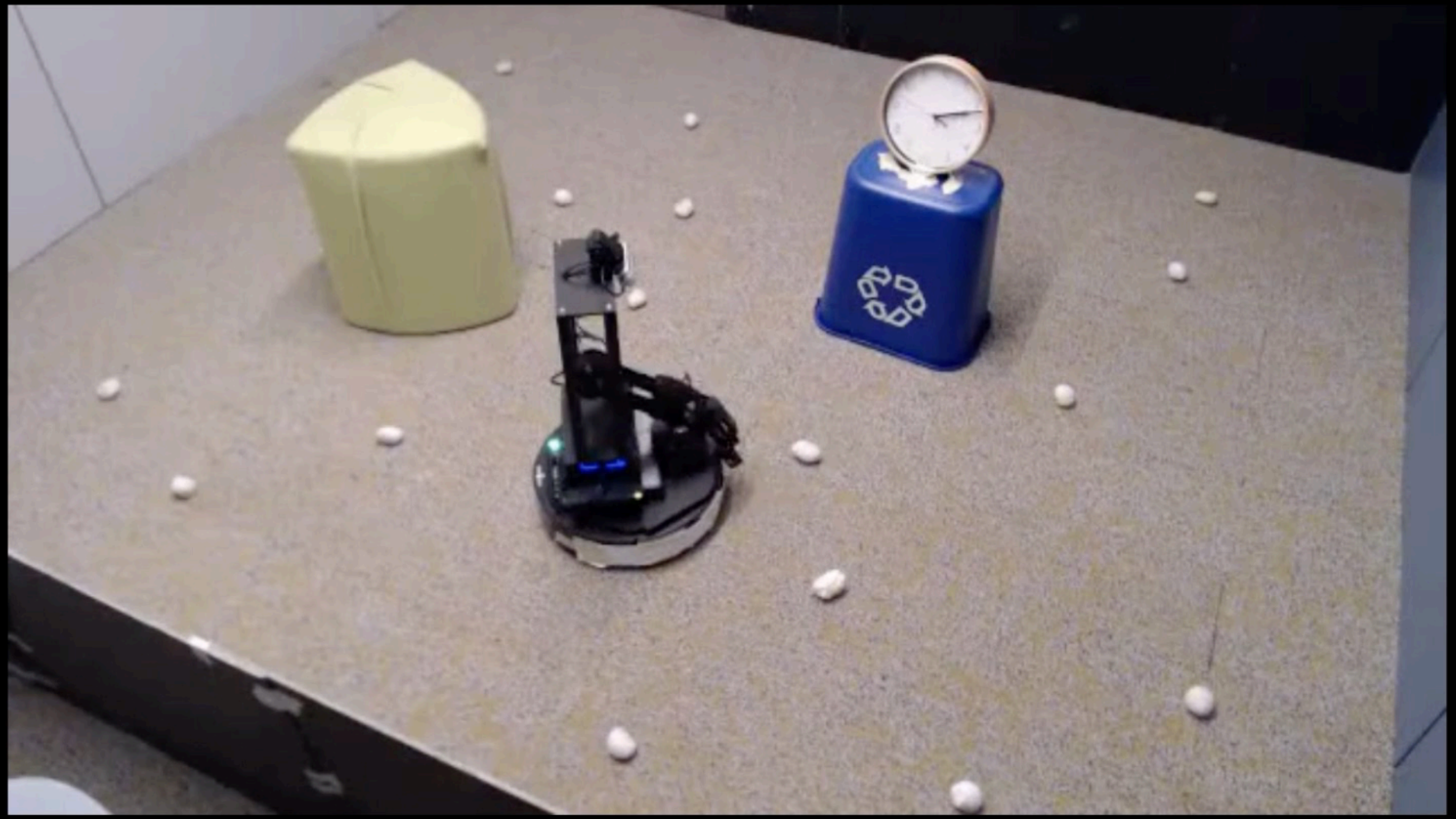


Algorithm 2 ReLMM (with Stationary Curriculum)

Hyperparameters: M, N_{st}, N_{grasp}, *relabel*
Init: function estimators π_n, $G^1, \ldots, G^M$.
Replay buffers $\mathcal{D}_n = \{\}$, $\mathcal{D}_g = \{\}$
TrainGrasp($G^1, .., G^M$, D_g, N_{st}, True)
for $t = 0, \ldots, T$ steps **do**
  Get navigation observation o_t
  Sample $a_n \sim \pi_n(\cdot|o_t)$ and perform a_n
  if *uniform*() $\leq \mathbb{P}[\text{grasp}|o_t]$ **then**
    r_g =TrainGrasp(G^1,.., G^M,D_g,N_{grasp},False)
  else $r_g = 0$
  if *relabel* **then**
    Navigation reward $r_n = \mathbb{P}[\text{grasp}|o_t] - 1$
  else $r_n = r_g - 1$
  Get next navigation observation o_{t+1}
  Store (o_t, a_n, r_n, o_{t+1}) in $\mathcal{D}_n$.
  Update π_n with $\mathcal{D}_n$ using SAC.
  Pseudo-reset
end for

1. perform a navigation action
2. if a grasp is likely to be successful, try that grasp
3. reward navigation based on either grasp success or (called relabel here) expected grasp success
4. update policy through soft actor critic (SAC)

ReLMM enables a robot to learn to cleanup a room without instrumentation or human supervision.



20x

The start of training in the no obstacle environment.

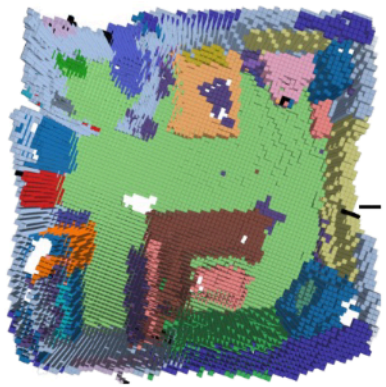
Summary: System-Level SSL

Illustrations by Matt Dixon

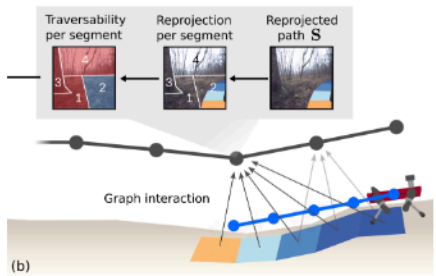
What if data pairs are collected together?

On a **system level**, self-supervised can be interpreted as learning from all available data of the system, but without any external input or labelling.

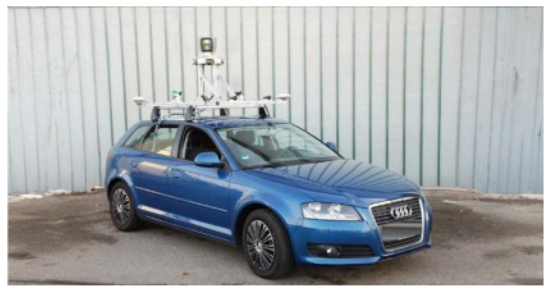
SSL from Paired Modalities



Pose + RGB + Depth



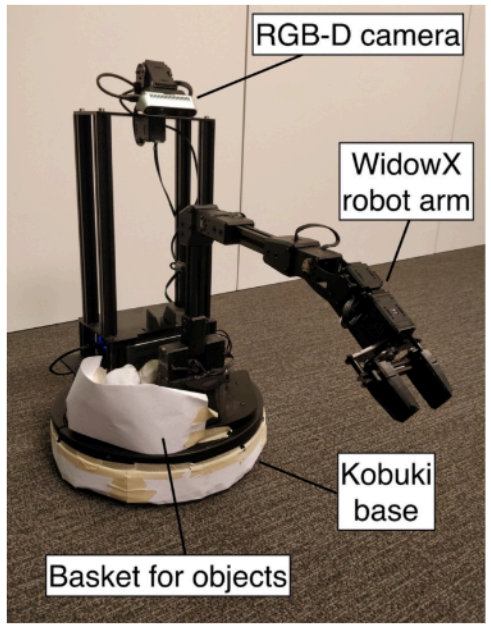
Pose + RGB + Legs



RGB + … -> Audio

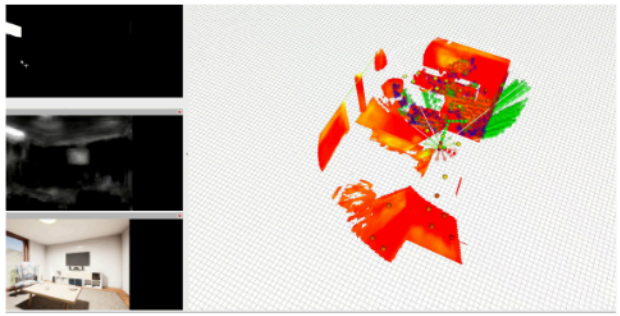
SSL from Physical Laws

enforcing consistency between different outputs



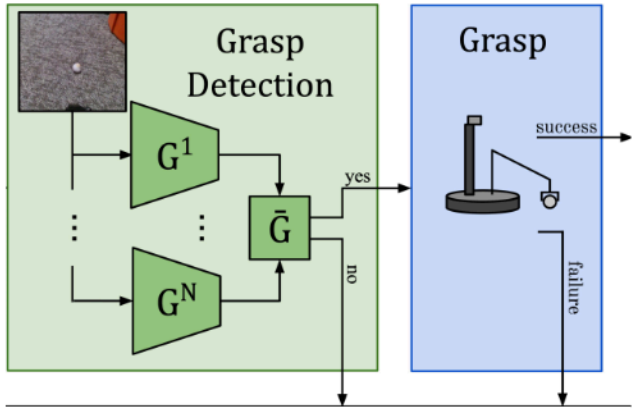
dropping is easier than grasping and generates randomness

In absence of curated data, uncertainty estimation is important.



uncertainty to select where to go

uncertainty to identify unknown



uncertainty to select informative actions

Next week

guest lecture on LLM training

information on master thesises in RPL